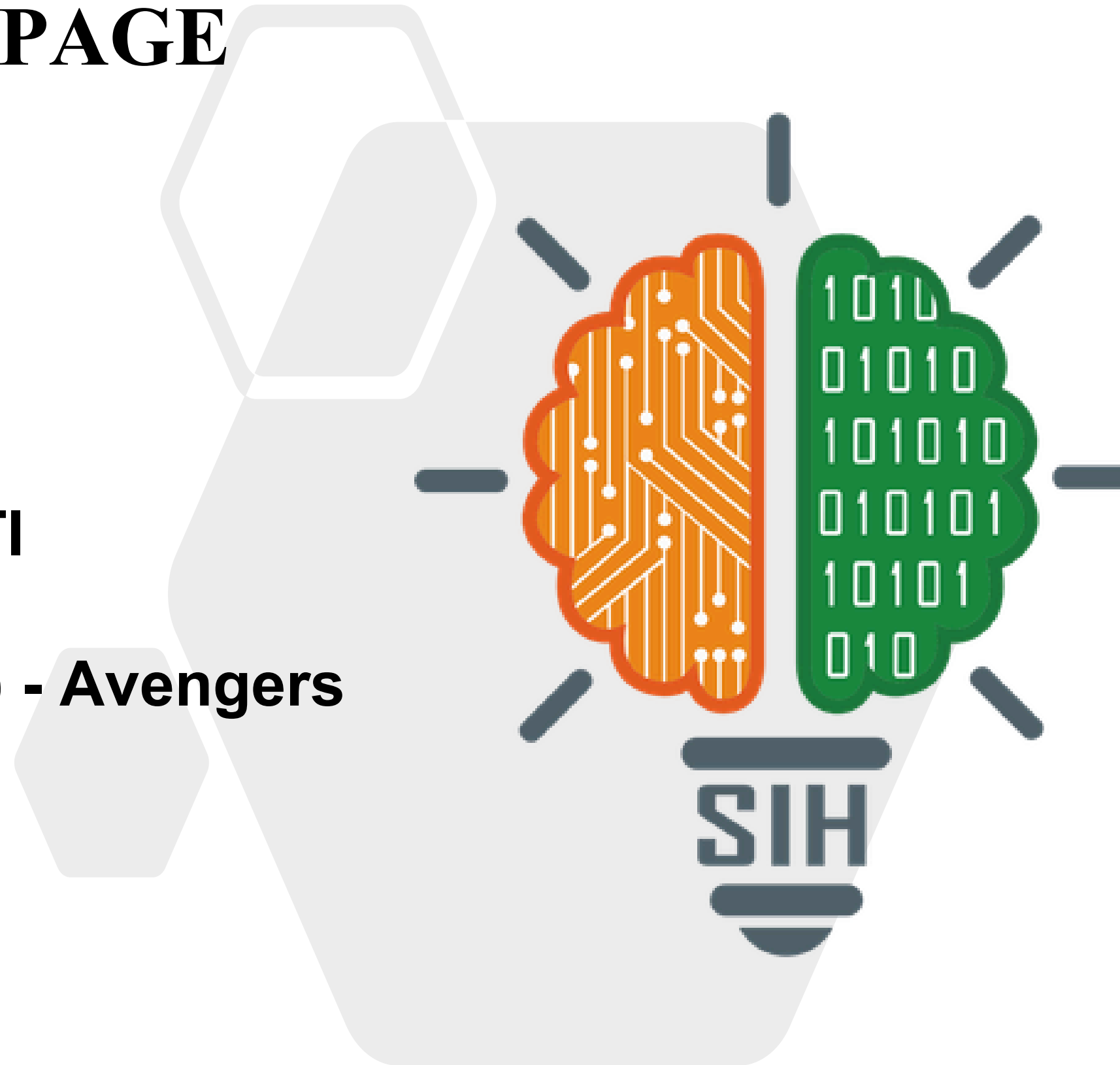


TITLE PAGE

- **Problem Statement ID - 26045**
- **Problem Statement Title - IP-SAKTI**
- **Team Name (Registered on portal) - Avengers**



⚠ THE PROBLEM

Ayurveda innovation sits inside a maze of overlapping IP, regulatory and biodiversity rules.

The cost of getting the answer wrong is not theoretical: traditional knowledge has already been targeted by misdirected patents, while practitioners and innovators must navigate fragmented national and international regimes.

7.85L registered AYUSH practitioners in India

3.72L registered Ayurveda practitioners

324+ TKDL-linked patent applications set aside / amended

Evidence: Ministry of Ayush, Approved Ayush in India 2024; TKDL, About TKDL. | Accessed Sep 2026.

💡 OUR SOLUTION

IP-SAKTI Sahayak

A multilingual, source-cited AI assistant that finds the right legal/regulatory evidence before it generates an answer.

- 1

CLASSIFY
formulation + intent
- 2

ROUTE
jurisdiction + IP domain
- 3

FILTER
metadata pre-filter
- 4

RETRIEVE
semantic + rerank
- 5

VERIFY
multi-source evidence
- 6

ANSWER
citations + confidence

Innovation: the same routing layer powers both speed and correctness. Qdrant searches only the metadata-matched slice of the corpus instead of treating every query as a global search.

✅ What changes with IP-SAKTI?

Current pain	IP-SAKTI mechanism	Result
Fragmented laws & rules	Jurisdiction + IP-domain routing	Right corpus first
Global search is expensive	Metadata pre-filter → vector search	Smaller search space
Generic AI answers	Section-level evidence + citations	Traceable answers
Traditional knowledge is hard to search	Multilingual embeddings + TKDL pointers	Better prior-art discovery
Law changes over time	Versioned source corpus	Current evidence



FEASIBILITY ANALYSIS



TECHNICAL FEASIBILITY

- 1 Modular RAG architecture with metadata-based query routing
- 2 Uses BGE-M3, Qdrant, FastAPI and LLMs for retrieval and synthesis
- 3 Progressive filtering and semantic search improve efficiency and relevance



ECONOMIC FEASIBILITY

- 1 Self-hosted embeddings reduce recurring API dependency
- 2 Remote storage and cache minimize local infrastructure requirements
- 3 Replaceable LLM, storage and cache providers enable cost flexibility



OPERATIONAL FEASIBILITY

- 1 Cloud-ready deployment using FastAPI, R2, Qdrant and Redis
- 2 Versioned authoritative sources enable reliable updating and maintenance
- 3 Confidence, abstention and human escalation support safe real-world use

UNIQUE SELLING POINT

Structure Aware RAG for Ayurveda IP & Regulatory Guidance



Jurisdiction-aware
Clear separation of India vs. International regimes



Domain-aware
Covers patents, GI, trademarks, copyright, designs, ABS and more



Formulation-aware
Classifies Ayurvedic products before legal reasoning



Evidence-grounded
Section-level citations with source verification



Safe-by-design
Confidence indicators, abstention and human escalation



**TRUSTED, TRACEABLE,
TAILORED FOR AYURVEDA**

THE REACH

7.85 lakh

registered AYUSH practitioners
in India

Practitioners

Fast, plain-language
regulatory guidance

Researchers

Traceable prior art
and authoritative
sources

AYUSH startups and MSMEs

IP strategy +
compliance
pathways

Cultivators and communities

GI / TK / ABS awareness

IP facilitators

Evidence-backed
escalation and audit
trail

VALUE CREATED ACROSS THE ECOSYSTEM

SOCIAL

Access to authoritative guidance

Reduces the knowledge gap between
legal / regulatory systems and AYUSH
practitioners.

ECONOMIC

Protect innovation and commercialise it

Helps identify relevant IP routes,
compliance duties and market-access
requirements earlier.

KNOWLEDGE

Defend traditional knowledge

Connects users to TKDL / prior-art
evidence and makes the protection
pathway actionable.

FROM QUESTION TO ACTION

ASK

User query



FIND

Relevant regime



VERIFY

Authoritative evidence



ACT

Next-step guidance

ANALYSIS



Search time



Traceability



Source confidence



Hallucination risk

Evidence base: Ministry of Ayush, Approved Ayush in India 2024; TKDL, About TKDL. Impact statements describe intended system-level benefits, not clinical or legal outcomes.